



OPEN Pharm D student's Knowledge, perception, and practice of CHAT-GPT in clinical training: a web-based cross-sectional survey in India

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The emergence of Large Language Models (LLMs) like ChatGPT captured significant attention in healthcare and pharmacy education, yet their real applicability and validity require extensive investigation in practice. Our study aimed to assess Pharm-D students' knowledge, perceptions, and practices (KPP) toward ChatGPT in clinical pharmacy practice training in India. A nationwide, web-based, cross-sectional survey was conducted between June and September 2025 to assess KPP towards ChatGPT use among PharmD students. A self-administered, pre-designed, and validated questionnaire was utilized to collect demographics, educational profiles, and KPP toward ChatGPT use in clinical training. We used social media platforms and Messenger applications and approached professional groups to recruit the study participants by using the snowball sampling technique. A chi-square test was applied to elucidate factors associated with KPP of ChatGPT use in the clinical training. The two-tailed P-value less than 0.05 was considered as a statistically significant value. In our study, the majority (>90%) of the PharmD students answered all basic knowledge questions about ChatGPT, except the source of ChatGPT's knowledge. The majority (85.8%) perceived that they indeed benefited from the use of ChatGPT in their training; the agreement levels for benefits varies across different activities. Regarding concerns, ChatGPT use can reduce the interaction with mentors (74.4%), increase the similarity index/AI detection score (64.1%), and increase the risk of inaccurate or misleading information (66.1%), ethical issues (63.3%), and limited applicability (61.0%). The use of ChatGPT in the selection of medicine and calculation of dose was low compared with other clinical pharmacy activities. Variables such as previous experience using AI tools and prior training on AI tool use were significantly associated with greater practice of ChatGPT. Conversely, the perceived benefits of ChatGPT were significantly positively associated with students who had completed internships ($P=0.019$) and had received prior training ($P=0.022$) on AI tools. There was a significant positive weak correlation between knowledge and perceived concerns ($r=0.177$; $P<0.001$), and moderate positive correlation between perceived benefits and practices of the participants ($r=0.377$; $P<0.001$) towards

ChatGPT use. Perceived concerns ($r = -0.126$; $P < 0.012$) were significantly negatively weakly correlated with ChatGPT use practices in clinical training. Majority of the PharmD students have good knowledge about ChatGPT, and positive perception towards ChatGPT benefits in clinical training. However, the practice was limited to certain activities due to concerns about accuracy, ethics, and reduced mentor interaction upon use of ChatGPT. Prior experience and training were significantly associated with high practice and positive attitudes. The study recommends to integrate AI tool use and application in pharmacy curriculum and train the faculty to promote competent and responsible use of ChatGPT in clinical pharmacy practice. Students must validate the information from the standard resources or interacting with mentors to deal with misleading information generated by AI tools.

Keywords Pharm d students, Knowledge, Perception, Practice, ChatGPT, Artificial intelligence, Clinical training, India

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The integration of artificial intelligence (AI) into healthcare has significantly transformed clinical practice and education, with voice assistants like Siri and Alexa as well as chat-based AI tools like ChatGPT being integral to modern communication and problem-solving¹. These tools work by simulating human cognition and assisting in planning and performing tasks and decision-making in real time². The ability of AI tools to drive personalized and interactive learning can increase the value of their use in education and clinical training³. Chat-Generative Pre-trained Transformer (Chat-GPT) is a free Large Language Model (LLM) developed by OpenAI (San Francisco, California, USA), in November 2022⁴. ChatGPT performs tasks such as answering questions, generating text, and simulating human like conversations. Chat-GPT became the rapid-growing consumer internet app that gained 100 million users within two months of its release⁵.

ChatGPT is a sensational innovation and a large user base that was used by the general public and healthcare professionals for medical information⁶. In the healthcare sector, ChatGPT represents an emerging platform for innovative perspectives, offering a wide range of applications including patient education, designing and conducting clinical studies, remote patient monitoring, accessing health information, scheduling medical appointments, symptom checkers for patients, medication reminders, developing tailored education programs for patients, digital assistants to help doctors improve patient care, enhancing communication within and between different medical specialties, improving healthcare providers' responses to insurance claims, providing telehealth services for underserved populations, and supporting mental health diagnosis and treatment sessions⁷. Additionally, ChatGPT assisting in radiological decision making, risk prediction and drug discovery process^{8,9}.

The rapid pace developments in AI technologies may greatly affects the healthcare systems. The appropriate use of digital health technologies can bring positive impact on patient care and treatment outcomes. The American Society of Health System Pharmacists (ASHP) highlights, in its "Statement on the Use of Artificial Intelligence in Pharmacy," that AI can enhance clinical decision support systems and improve patient care; moreover, it can optimise inventory management and product verification within pharmacy operations¹⁰. These statements suggest that the development of digital technology use in the pharmacy field should commence early without delay to have positive impact on patient care. Despite an interest in use and development of AI in healthcare has increased, their acceptance and feasibility in medical and pharmacy education were insufficiently explored¹¹.

In the pharmacy field, the use of AI enhances the pharmacists job efficiency in activities such as drug interaction screening, prescription reconciliation, and patient involvement¹². Evidence revealed that ChatGPT assists in drug counselling, review of prescriptions, and patient education¹³. However, a comparative study between ChatGPT versus clinical pharmacist revealed that ChatGPT excels in drug counselling, but show less performance in ADR detection, prescription review and patient education¹⁴. Clinical pharmacist plays a critical role in providing pharmaceutical care services to achieve definite clinical outcomes and improve the drug safety by minimizing drug induced harm¹⁵. The integration of AI tools in clinical pharmacy practice training and practice can augment the clinical decision making, and patient outcomes¹⁶.

Though the previous studies have shown promising findings for the use of ChatGPT in the healthcare sector, but the accuracy and reliability need to be explored in the real-world scenario. Few studies demonstrated the importance of ChatGPT in evaluating diverse clinical scenarios to validate its practical application^{13,17,18}. Additionally, there are diverse methodologies to investigate the accuracy of ChatGPT in clinical pharmacy training^{14,19}. One of the notables inquire is employing 20 pharmacotherapeutic case in determining accuracy and consistency of the ChatGPT¹⁹. These findings revealed that there were wide variations in the accuracy rate across different temporal intervals¹⁹.

Recent research evidence revealed ChatGPT's medical education potential, ranging from assisting students in easy understanding of complex topics to providing tutorials and aiding assignments²⁰. The integration of ChatGPT in clinical training of Pharm D students remains to be investigated in Indian context. Based on current literature, a significant gap exists in connection to the use of Large Language Models (LLM) such as ChatGPT in clinical training by the Pharm D students in the India. This study aimed to assess Pharm-D students' perceptions, concerns, and practices of integrating ChatGPT into pharmacy practice training. The findings of the current study will underscore the transformative potential of ChatGPT in clinical training of practicing Pharm D students. Additionally, these findings assist education councils and universities to develop guidelines for the use of LLM such as ChatGPT in the clinical training of pharmacists.

Materials and methods

Study design and ethical considerations

A web-based, cross-sectional, online survey was conducted among Indian practicing Pharm D students for four months from June to September 2025. The Sri Padmavathi School of Pharmacy (SPSP/24–25/PHD20) located in Tirupati approved the research proposal, and informed consent process. Online informed consent was obtained from all participants involved in this survey. No monetary payment was given for participation, and anonymity of the identity was maintained to ensure the confidentiality and reliability of the data. This online survey followed the ethical principles outlined in the Declaration of Helsinki for research involving human participants and conducted according to the STROBE guidelines for reporting of observational studies.

Survey participants

The survey included students enrolled in Pharm D (Doctor of Pharmacy) program in any private or government institute or university located in India and practicing clinical training in the hospital as a part of their clerkship (Pharm D V Year) or internship (Pharm D V Year) according to the Pharmacy Council of India (PCI) academic regulations. Before responding to the online survey questions, participants are provided access to the information regarding purpose, targeted audience, estimated time, confidentiality, nature of participation, objectives, and outcomes of the study that are displayed on the first page. Before responding to the survey questions, participants must read the information and select “yes” for the first three questions in the online google form: (1) Are you practicing Pharm D (Internship or clerkship) student, (2) Have you ever used ChatGPT in your clinical training, and (3) Are you willing to join in this online survey.

Sample size determination and sampling procedure

The number of practicing Pharm D students required to participate in this study was calculated using the single-population proportion formula ($n = Z^2 p(1-p)/d^2$), assuming the population is infinite or $\geq 10,000$. Since there was no previous study in the Indian context, we assumed 50% (p) of Pharm D students were using ChatGPT in their clinical training, margin of error (d) 5%, design effect 1%, the power of the study 80%, the Z-value at 95% confidence level is 1.96, and the final sample size was calculated as 384. The calculation was as follows: $n = (1.96)^2 (0.5) (0.5) / (0.05)^2 = 384$. A 10% of the sample size was added to cater for the non-response rate, and the final sample size was estimated as 422.

A non-probability sampling method like snow-ball sampling was used to capture the required participants. Initially, the survey link was disseminated to institutional and departmental heads and requested to post in the online Pharm D student portal and social media platforms (WhatsApp, LinkedIn, IMO, and Instagram). All invited participants are requested to share the survey link to their familiar groups and contacts to obtain required sample size. Though the snow-ball method of sampling has limitations in terms of generalizability of the findings, however it provides maximal sample size as per study requirement.

Survey tool

The developed survey tool comprises five sections: (1) Demographics, academics, and AI tool usage profile; (2) Knowledge about ChatGPT; (3) Perceived benefits and concerns of ChatGPT in Clinical Training; and (4) Practice of ChatGPT in Clinical Training.

Demographics, academics, and AI tool usage profile

The demographic profile such participants age, gender, residence and state were included in this section. Academic section comprises year of study, current percentage or GPA, name of the institute or university, and completion status of internship training. This section also covers previous use of AI tools in the clinical practice, and training on use of AI tools in clinical practice.

Knowledge about ChatGPT

The knowledge section consists 10 questions regarding ChatGPT in connection to AI chatbot form, version, developed by whom, function, features availability, source of knowledge, requirement of internet connection for operation, and generation of human text like conversations. Knowledge questionnaire comprises questions having both dichotomous response (True/False or Yes/No) and multiple options (one option is correct answer). Participants response to each question was scored as one point for correct answer, and zero for wrong answer, which acquires maximum of 10 and minimum of zero for all knowledge questions. After estimation of Knowledge scores for each participant, Bloom's cut-off scale was used to divide knowledge levels into good (80–100% correct answer; score 8–10), moderate (60–79% correct answer; score 6–7), and poor (<60% correct answer; score <6) knowledge about ChatGPT among practicing Pharm D students.

Perceived benefits and concerns of ChatGPT in clinical training

The perception of practicing Pharm D students was divided into perceived benefits and perceived concerns about ChatGPT use in clinical training. The perceived benefits of the use of ChatGPT in clinical training of Pharm D students were assessed by using 16 statements. The benefits of ChatGPT such as assists in clinical training, promotes rational drug use and improves patient outcomes, identify medication related problems (MRPs), dose calculation, develops educational content for patient counselling, remind medication intake to improve adherence, identification of drugs that need to be added to the treatment regimen, identify wrong drugs present in the prescription, provide evidence-based drug information to HCPs, aids in the development of pharmaceutical care plan, summarize treatment guidelines, documentation of clinical interventions, and assist students in writing clinical reports as per university guidelines were included in this section. Each statement in the perceived benefits section was graded on a 5-point Likert scale, with negative and positive responses such as strongly disagree 1, disagree 2, neutral 3, agree 4, and strongly agree 5. The maximum expected score from each participant for the perceived benefits domain was 80, and the minimum score was 16. After estimation of perceived benefits scores for each participant, Bloom's cut-off scale was used to divide perception of Pharm D students about ChatGPT benefits into high (80–100% positive response; score 64–80), neutral (60–79% positive response; score 48–63), and low (<60% positive response; score <48).

The perceived concerns regarding ChatGPT use in clinical training of Pharm D students were assessed by using 10 statements. Perceived concerns such as ChatGPT use may results in lack of personal interaction with mentors, affects critical thinking and problem solving skills, generate inaccurate and misleading response, affects privacy and confidentiality of the patient's information, not accessible due to lack of resources, limited scope and applicability, cause underutilization of the essential information resources, produce inconsistent responses across similar queries, cause risk of bias in response, and increase the similarity index and AI detection score in clinical training reports were included in this section. Each statement in the perceived concerns section was graded on a 5-point Likert scale, with negative and positive responses such as strongly disagree 1, disagree 2, neutral 3, agree 4, and strongly agree 5. The maximum expected score from each participant for the perceived concerns domain was 50, and the minimum score was 10. After estimation of perceived concerns scores for each participant, Bloom's cut-off scale was used to divide perception of Pharm D students about ChatGPT concern into high (80–100% positive response; score 40–50), neutral (60–79% positive response; score 30–39), and low (<60% positive response; score <30).

Practice of ChatGPT in clinical training

The practice of ChatGPT in Clinical Training among Pharm D students was assessed by using 10 questions having dichotomous responses (Yes/No). The practice section comprises questions regarding ChatGPT used in clinical training, screening drug interaction, identifying and managing ADRs, calculation of dose or dosage regimen, to respond drug or poison information query, selection of drug in the treatment of disease, to obtain counselling tips, development and implementation of pharmaceutical care plan, to write SOAP notes or clinical interventions or report writing, and recommending to other Pharm D students or preceptors to use in clinical training. Each question response "yes" was weighted as one, "no" as zero. The maximum expected score from each participant for the practice of ChatGPT in clinical training was 10, and the minimum score was zero. After estimation of practice scores for each participant, Bloom's cut-off scale was used to divide Pharm D students' practice of ChatGPT in clinical training into high (80–100% responses were yes; score 8–10), moderate (60–79% responses were yes; score 6–7), and low (<60% responses were yes; score <6).

Development and validation of a survey tool

A self-administered survey tool was constructed based on the past literature on ChatGPT and AI tools utility among practicing pharmacists or pharmacy or medical students^{1,13,18,21–25}. The developed survey tool was subjected to content validity and reliability assessment. The content present in the survey tool was validated by a panel of experts composed with Pharm D professor (1), preceptor of Pharm D students (1), clinical pharmacist (1), AI tool specialist (1), and clinical specialist (1). About 46 questions/statements (knowledge = 10; perceived benefits = 16; perceived concerns = 10; practice = 10) were present in the survey tool to understand ChatGPT use in clinical training. The expert response to include each question or statement in the survey tool was rated on a 4-point Likert scale that ranges between strongly agree (4) and strongly disagree (1). After rating, content validity index indicators (S-CVI per average number and S-CVI per universal agreement) were determined for knowledge (0.96, 0.80), perceived benefits (0.96, 0.81), perceived concerns (0.94, 0.80), and practice (0.92, 0.80) domains of the survey tool. S-CVI indicators for all domains are within the acceptable range (≥ 0.8).

An offline pilot survey was conducted among 30 practicing Pharm D students to determine the internal consistency of the survey tool. The findings of the participants included in the pilot study were excluded from the final survey. Cronbach's values of knowledge (Cronbach's Alpha = 0.867), perceived benefits (Cronbach's Alpha = 0.803), perceived concerns (Cronbach's Alpha = 0.827), and practice (Cronbach's Alpha = 0.855) domains showed an acceptable internal consistency.

Data collection

The data was collected by sending google survey link to Pharm D V- and VI-year students through various messenger (WhatsApp, IMO, WeChat) and social media (LinkedIn, Facebook) groups. All students are requested to provide online consent to participate in the survey. Personal request was made through phone call to university or institutional Pharm D teaching staff to motivate their Pharm D students to participate in the online survey. Background information about the survey was also provided to teaching staff to guide the interested students to fill the online survey form.

Data analysis

Upon collecting data from the practicing Pharm D students, IBM SPSS statistics for windows, version 25 (IBM Corp., Armonk, NY, USA) was used for analysis. Descriptive statistics were used to present the demographics, academic profile, AI tool usage profile, and KPP of the ChatGPT use among practicing Pharm D students. Normally distributed data was presented using mean and standard deviation while variables that were not normally distributed were described using median and interquartile range. A Pearson correlation analysis was performed to assess the association KPP scores of practicing Pharm D students toward ChatGPT use in clinical training. Bivariate and multivariate logistic regression analysis was applied to determine the factors associated with good knowledge, good perception, high concern, and high practice of the ChatGPT use in clinical training of Pharm D students. Factors with a p value less than 0.20 at bivariate level was considered at multivariate level. Variables with a p value less than 0.05 were statistically significant. The significant associated factors were presented with their adjusted odds ratio and corresponding 95% confidence intervals.

Results

In this online survey, about 407 practicing PharmD students represented from 49 institutions and 12 states. Out of the 407 students, 12 were removed due to unwillingness, leaving 395 students for data analysis.

Demographics, academic profile, and AI tool usage of the study participants

Our study findings revealed that majority of the respondents were aged less than or equal to 23 years (mean 22.93 and SD 1.68), female (71.9%), and from the state of Andhra Pradesh (72.7%). In our study, the residence of the participants shares nearly equal from rural (40.5%) and urban (45.8%) regions. Regarding academics, more than three-fourth of the respondents were studying in private institutions / universities (87.3%), and attained above 7 GPA (73.9%). Majority (70.4%) of the students were not yet completed their internship and more than half (59.2%) of respondents were pursuing Pharm D VI year. The AI tool findings revealed that very few students (29.6%) were undergone training regarding AI tool use in clinical training and more than half of the respondents (56.5%) had previous experience of use of AI tool. The demographics, academics, and AI tool use characteristics of the participants were presented in Table 1.

Knowledge about ChatGPT use among practicing pharm D students

In our study, majority of students are aware about ChatGPT is an AI chatbot (95.7%), its function (96.2%), availability of plus and pro versions with an advanced feature (96.2%), ChatGPT runs with internet connection (99.5%), available in multiple languages (93.4%), and it can generate human like text conversations (96.7%). Regarding source of ChatGPT's knowledge, only 32.4% of the students correctly responded. However, a considerable proportion (70.1%) of the students were aware about the latest version of ChatGPT. The distribution of the responses of Pharm D students to knowledge questions were presented in Table 2.

Perception towards ChatGPT use in clinical training among pharm D students

Perceived benefits

Majority (85.8%) of the practicing Pharm D students perceived that they indeed benefit from the use of ChatGPT in their clinical training. Most of the respondents perceived that ChatGPT enhances the PharmD students' documentation skills (79.2%), provides information to identify and manage ADRs, and promotes rational drug use and improves patient outcomes (74.4%). The distribution of the perceived benefits of ChatGPT use in clinical training among PharmD students was presented in Table 3.

Perceived concerns

Parallel to the perceived benefits, respondents also agreed with several potential concerns regarding the use of ChatGPT in clinical training. Nearly three-fourth of the respondents were concern about the use of ChatGPT may reduce the interaction with mentors. More than half of the respondents have concerns regarding increased similarity index / AI-detection score (64.1%), provides inaccurate or misleading information (66.1%), and affects privacy and confidentiality of the patient information (63.3%). The distribution of the perceived concerns of ChatGPT use in Clinical Training among Pharm D students were presented in Table 3.

Practice of ChatGPT use in clinical training among pharm D students

In our study, majority of the PharmD students are recommending (89.8%) ChatGPT to their fellow students and preceptors, and using to get counselling tips to provide patient specific counselling session. About 80.3% of the respondents were regular using the ChatGPT in their clinical training and 73.4% were utilizing in providing drug and poison information. The distribution of the practice of ChatGPT use in clinical training of Pharm D students were presented in Table 4.

Adequacy of knowledge, perception, and practice of ChatGPT use in clinical training by practicing pharm D students

In our study, the majority of the respondents have good knowledge regarding ChatGPT, but this is non-specific to clinical training. Regarding perception level, half of the respondents perceived neutral (50.6%), followed by high (45.1%), towards the benefits of ChatGPT in clinical pharmacy training. A significant proportion (73.7%) of the respondents were neutral regarding perceived concerns about ChatGPT use in clinical training. However, three-fourths of the students were concerned that ChatGPT may have an impact on interaction with the mentor. More than half (57.2%) of the respondents were practicing the use of ChatGPT in the clinical pharmacy training. The distribution of the adequacy of knowledge, perception, and practice of ChatGPT use in training was presented in Table 5. The findings of correlation analysis revealed that respondents' knowledge score ($r=0.177$; $P<0.001$) was

Variable	Frequency (%)
Age	22.93 ± 1.68
≤ 23	290 (73.4)
> 23	105 (26.6)
Gender	
Male	111 (28.1)
Female	284 (71.9)
Residence	
Rural	160 (40.5)
Semi-urban	54 (13.7)
Urban	181 (45.8)
State	
Andhra Pradesh	287 (72.7)
Arunachal Pradesh	1 (0.3)
Assam	2 (0.5)
Gujarat	1 (0.3)
Karnataka	20 (5.1)
Kerala	13 (3.3)
Madhya Pradesh	1 (0.3)
Maharashtra	2 (0.5)
Mizoram	1 (0.3)
Tamil Nadu	30 (7.6)
Telangana	36 (9.1)
West Bengal	1 (0.3)
Current year of the study	
Pharm D V Year	161 (40.8)
Pharm D VI Year	234 (59.2)
Overall current GPA or percentage	
≤ 7	103 (26.1)
> 7	292 (73.9)
Institute / university type	
Government	50 (12.7)
Private	345 (87.3)
Did you complete your internship training in the hospital	
Yes	117 (29.6)
No	278 (70.4)
Do you have any previous experience with the use of AI tools?	
Yes	223 (56.5)
No	172 (43.5)
Have you ever received training on how to use AI tools in clinical training?	
Yes	90 (22.8)
No	305 (77.2)

Table 1. Demographics, academic profile, and AI tool usage of the study participants.

positively weakly correlated with the perceived concerns score of ChatGPT use in clinical training. Respondents perceived benefits score ($r = 0.377$; $P < 0.001$) was moderately positively correlated, and the perceived concerns score ($r = -0.126$; $P < 0.012$) was weakly negatively correlated with the practices of ChatGPT use in the clinical training.

Association of practicing pharm D students' profile with their perception and practice of ChatGPT use in clinical training

In our study, students who completed internship training showed a significant association ($\chi^2 = 7.906$; $p = 0.019$) with the perceived benefits offered by ChatGPT in clinical training compared to those who had not. There was a significant association ($\chi^2 = 15.012$; $p = 0.001$) between previous experience in using AI tools and higher practice of ChatGPT in clinical training. Prior training on AI tools use in clinical training showed a significant

Knowledge question	Correct (%)
K1. What is ChatGPT	310 (78.5)
K2. ChatGPT is an advanced form of AI chatbots	378 (95.7)
K3. Who developed ChatGPT?	339 (85.8)
K4. What is the latest version of ChatGPT?	277 (70.1)
K5. What is the main function of ChatGPT?	380 (96.2)
K6. ChatGPT Plus and Pro are the paid versions and have advanced features	380 (96.2)
K7. What is the source of ChatGPT's knowledge?	128 (32.4)
K8. Does ChatGPT required internet connection?	393 (99.5)
K9. Does ChatGPT works in multiple languages?	369 (93.4)
K10. Does ChatGPT generates human like text conversations?	382 (96.7)

Table 2. Knowledge about ChatGPT use among practicing pharm D students.

Statement	SA/A (%)	N (%)	D/SD (%)
Perceived Benefits			
PB1. A practicing PharmD student can benefit from using ChatGPT in clinical training	339 (85.8)	50 (12.7)	6 (1.5)
PB15. ChatGPT enhances the documentation skills of the PharmD students in writing SOAP notes and clinical interventions	313 (79.2)	67 (17.0)	15 (3.8)
PB5. ChatGPT provides information to identify and manage ADRs	306 (77.5)	74 (18.7)	15 (3.8)
PB2. Use of ChatGPT in clinical training promotes rational drug use and improves patient outcomes	294 (74.4)	82 (20.8)	19 (4.8)
PB3. ChatGPT helps in identifying and resolving medication-related problems (MRPs)	293 (74.2)	86 (21.8)	16 (4.1)
PB13. ChatGPT aids in developing a pharmaceutical care plan.	292 (73.9)	89 (22.5)	14 (3.5)
PB14. ChatGPT can summarize the up-to-date treatment guidelines.	291 (73.7)	93 (23.5)	11 (2.8)
PB7. ChatGPT assists in providing tailored and unbiased educational content about medications and diseases while counselling the patient	289 (73.2)	88 (22.3)	18 (4.6)
PB4. Drug-drug interactions can be easily screened and managed by using ChatGPT	283 (71.6)	96 (24.3)	16 (4.1)
PB8. ChatGPT helps in designing patient-specific reminders to improve medication adherence	278 (70.4)	94 (23.8)	23 (5.8)
PB6. ChatGPT helps in calculating the dose and dosage regimen as per the patient's requirement	276 (69.9)	87 (22.0)	32 (8.1)
PB12. ChatGPT assists in providing appropriate, unbiased, evidence-based information about drugs and poisons to healthcare professionals	273 (69.1)	109 (27.6)	13 (3.3)
PB10. ChatGPT provides information about medication(s) that need to be added to the patient's treatment	269 (68.1)	91 (23.0)	35 (8.9)
PB16. ChatGPT assists in writing clinical training internship reports as per university / institutional guidelines.	267 (67.6)	102 (25.8)	26 (6.6)
PB9. ChatGPT assists in screening for unnecessary medication in the patient's prescription	255 (64.6)	106 (26.8)	34 (8.6)
PB11. ChatGPT helps to identify the wrong medication(s) that are prescribed in the patient's treatment	254 (64.3)	120 (30.4)	21 (5.3)
Perceived concerns			
PC1. By the regular use of ChatGPT in clinical training, students may lack personal interaction with their mentors.	294 (74.4)	80 (20.3)	21 (5.3)
PC3. There is possibility that ChatGPT may generate inaccurate and misleading response due to its limitation in contextual understanding and lack of real-time verification.	261 (66.1)	100 (25.3)	34 (8.6)
PC10. The use of ChatGPT may increase the similarity index or AI writing detection score in clinical training internship report writing.	253 (64.1)	130 (32.9)	12 (3.0)
PC4. ChatGPT may raise ethical concerns such as privacy and confidentiality of the patient information	250 (63.3)	114 (28.9)	31 (7.8)
PC6. ChatGPT may not aid in all the necessary areas of clinical pharmacy training due to the limited scope and applicability.	241 (61.0)	138 (34.9)	16 (4.1)
PC9. There is a risk bias in ChatGPT responses or recommendations in clinical training	231 (58.5)	129 (32.7)	35 (8.9)
PC7. Overreliance on ChatGPT may lead students to underutilize essential information resources in the field of clinical pharmacy.	160 (40.5)	137 (34.7)	98 (24.8)
PC5. Some students may don't have access to use ChatGPT in their training due to lack of resources	158 (40.0)	123 (31.1)	114 (28.9)
PC2. The use of ChatGPT in clinical training may affect the students critical thinking and problem-solving skills	151 (38.2)	102 (25.8)	142 (35.9)
PC8. Inconsistency in responses across similar queries can reduce trust in the reliability of ChatGPT for clinical use	142 (35.9)	149 (37.7)	104 (26.3)

Table 3. Perceived benefits and concerns of ChatGPT use in clinical training among pharm D students.

association with perceived benefits ($\chi^2=7.611$; $p=0.022$) and greater practice ($\chi^2=18.105$; $p<0.001$) of ChatGPT in clinical training. The association of the student's profile with perception and practice of ChatGPT use in clinical training was presented in Table 6.

Practice question	Yes (%)	No (%)
P10. Did you recommend ChatGPT to other Pharm D students or their preceptors to use in clinical training?	331 (89.8)	64 (16.2)
P7. Have you ever used ChatGPT in getting counselling tips to provide patient specific counselling session?	330 (83.5)	65 (16.5)
P1. Have you ever used ChatGPT in clinical training?	317 (80.3)	78 (19.7)
P5. Have you ever used ChatGPT as an information resource to retrieve drug or poison information upon receiving a drug information query?	290 (73.4)	105 (26.6)
P2. Have you ever used ChatGPT in screening and identifying drug-drug interactions?	279 (70.6)	116 (29.4)
P3. Have you ever used ChatGPT in identifying and managing adverse drug reactions?	278 (70.4)	117 (29.6)
P8. Have you ever used ChatGPT in developing and implementing a pharmaceutical care plan?	266 (67.3)	129 (32.7)
P9. Have you ever used ChatGPT in writing SOAP notes and clinical interventions or report writing?	262 (66.3)	133 (33.7)
P6. Have you ever used ChatGPT in the selection of a suitable drug in disease management?	259 (65.6)	136 (34.4)
P4. Have you ever used ChatGPT to calculate the dose and dosage regimen for the patient?	231 (58.5)	164 (41.5)

Table 4. Practice of ChatGPT use in clinical training among pharm D students.

Adequacy level	Frequency (%)
Knowledge	
Good	321 (81.3)
Moderate	65 (16.5)
Poor	9 (2.3)
Level of perception	
High	178 (45.1)
Neutral	200 (50.6)
Low	17 (4.3)
Level of concern	
High	75 (19.0)
Neutral	291 (73.7)
Low	29 (7.3)
Level of practice	
High	226 (57.2)
Moderate	56 (14.2)
Low	113 (28.6)

Table 5. Adequacy of knowledge, perception, and practice of ChatGPT in clinical training by pharm D students.

Discussion

Application of ChatGPT in clinical training, education, and research is tremendously increased, and it has high potential to engage students and faculty in teaching and learning through creativity, and innovation with high efficiency and less clinical work load. For certain tasks, the capabilities of these tools exceed the limitations of humans. However, it showed friction in dealing tasks that were linked with human versatility, emotional intelligence, and complex intersecting phenomenon. The current study was conducted among practicing Pharm D students to assess their perceptions and practices of ChatGPT in their clinical training (clerkship and internship) that is scheduled in V and VI years as per curriculum stated by the Pharmacy Council of India (PCI).

Knowledge about ChatGPT

Our study findings revealed that the majority (95.7%) of the Pharm D students were aware that ChatGPT is an advanced AI chatbot. These findings were nearly similar with a study conducted among pharmacists (90.1%) in Saudi Arabia in identifying ChatGPT is an advanced chatbot. Though this study revealed high knowledge for the basic question, but the practicing pharmacists showed wide gaps in deeper knowledge levels, such as open-AI developed ChatGPT, availability of the free and paid versions, and ChatGPT can simulate human like conversation. In our study more than 90% of the Pharm D students are aware of these functions given by the Chat GPT. Additionally, our study participants revealed high knowledge towards function of ChatGPT, ChatGPT runs with internet connection, and it is availability in multiple languages. The high level of awareness among Pharm D students suggest that ChatGPT achieved high level of recognition and visibility due to its widespread use in clinical and academic settings, and increased digital literacy among students' community²⁶. In our study, very few Pharm D students (32.4%) are aware about the source of ChatGPT's knowledge. This is a very important and alarming finding that can implicate students to misuse or over-trust ChatGPT in clinical scenarios. A systematic review shows that low AI knowledge was reported among healthcare students from the studies conducted in low-and-middle income countries²⁷. The primary reason mentioned in this review for not having adequate knowledge and skills among healthcare students was no relevant training course or lack

Variable	Total (%)	Perception toward benefits			Perception toward concerns			Level of practice		
		High (%)	Neutral (%)	Low (%)	High (%)	Neutral (%)	Low (%)	High (%)	Mod. (%)	Low (%)
Age										
≤23	290 (73.4)	132 (45.5)	144 (49.7)	14 (4.8)	60 (20.7)	206 (71.0)	24 (8.3)	163 (56.2)	48 (16.6)	79 (27.2)
>23	105 (26.6)	46 (43.8)	56 (53.3)	3 (2.9)	15 (14.3)	85 (81.0)	5 (4.8)	63 (60.0)	8 (7.6)	34 (32.4)
		χ ² = 0.951; p-value = 0.621			χ ² = 3.991; p-value = 0.136			χ ² = 5.244; p-value = 0.073		
Gender										
Male	111 (28.1)	54 (48.6)	49 (44.1)	8 (7.2)	14 (12.6)	90 (81.1)	7 (6.3)	67 (60.4)	14 (12.6)	30 (27.0)
30Female	284 (71.9)	124 (43.7)	151 (53.2)	9 (3.2)	61 (21.5)	201 (70.8)	22 (7.7)	159 (56.0)	42 (14.8)	83 (29.2)
		χ ² = 4.748; p-value = 0.093			χ ² = 4.680; p-value = 0.096			χ ² = 0.668; p-value = 0.716		
Residence										
Rural	160 (40.5)	67 (41.9)	90 (56.3)	3 (1.9)	31 (19.4)	117 (73.1)	12 (7.5)	95 (59.4)	24 (15.0)	41 (25.6)
Semi-urban	54 (13.7)	20 (37.0)	30 (55.6)	4 (7.4)	10 (18.5)	41 (75.9)	3 (5.6)	27 (50.0)	11 (20.4)	16 (29.6)
Urban	181 (45.8)	91 (50.3)	80 (44.2)	10 (5.5)	34 (18.8)	133 (73.5)	14 (7.7)	104 (57.5)	21 (11.6)	56 (30.9)
		χ ² = 8.988; p-value = 0.061			χ ² = 0.346; p-value = 0.987			χ ² = 3.869; p-value = 0.424		
Current Year										
Pharm D V Year	161 (40.8)	80 (49.7)	75 (46.6)	6 (3.7)	33 (20.5)	114 (70.8)	14 (8.7)	96 (59.6)	22 (13.7)	43 (26.7)
Pharm D VI Year	234 (59.2)	98 (41.9)	125 (53.4)	11	42 (17.9)	177 (75.6)	15 (6.4)	130 (55.6)	34 (14.5)	70 (29.9)
		χ ² = 2.381; p-value = 0.304			χ ² = 1.307; p-value = 0.520			χ ² = 0.670; p-value = 0.716		
GPA										
≤7	103 (26.1)	38 (36.9)	60 (58.3)	5 (4.9)	17 (16.5)	74 (71.8)	12 (11.7)	49 (47.6)	19 (18.4)	35 (34.0)
>7	292 (73.9)	140 (47.9)	140 (47.9)	12 (4.1)	58 (19.9)	217 (74.3)	17 (5.8)	177 (60.6)	37 (12.7)	78 (26.7)
		χ ² = 3.760; p-value = 0.153			χ ² = 4.039; p-value = 0.133			χ ² = 5.462; p-value = 0.065		
Institute type										
Government	50 (12.7)	28 (56.0)	18 (36.0)	4 (8.0)	10 (20.0)	38 (76.0)	2 (4.0)	33 (66.0)	6 (12.0)	11 (22.0)
Private	345 (87.3)	150 (43.5)	182 (52.8)	13 (3.8)	65 (18.8)	253 (73.3)	27 (7.8)	193 (55.9)	50 (14.5)	102 (29.6)
		χ ² = 5.758; p-value = 0.056			χ ² = 0.944; p-value = 0.624					
Completion of internship training										
Yes	117 (29.6)	58 (49.6)	59 (50.4)	0 (0.0)	22 (18.8)	87 (74.4)	8 (6.8)	77 (65.8)	15 (12.8)	25 (21.4)
No	278 (70.4)	120 (43.2)	141 (50.7)	17 (6.1)	53 (19.1)	204 (73.4)	21 (7.6)	149 (53.6)	41 (14.7)	88 (31.7)
		χ ² = 7.906; p-value = 0.019			χ ² = 0.071; p-value = 0.965			χ ² = 5.409; p-value = 0.067		
Previous experience to use AI tools										
Yes	223 (56.5)	106 (47.5)	106 (47.5)	11 (4.9)	39 (17.5)	171 (76.7)	13 (5.8)	144 (64.6)	32 (14.3)	47 (21.1)
No	172 (43.5)	72 (41.9)	94 (54.7)	6 (3.5)	36 (20.9)	120 (69.8)	16 (9.3)	82 (47.7)	24 (14.0)	66 (38.4)
		χ ² = 2.136; p-value = 0.344			χ ² = 2.831; p-value = 0.243			χ ² = 15.012; p-value = 0.001		
Trained to use AI tools in clinical training										
Yes	90 (22.8)	52 (57.8)	35 (38.9)	3 (3.3)	14 (15.6)	72 (19.0)	4 (4.4)	68 (75.6)	11 (12.2)	11 (12.2)
No	305 (77.2)	126 (41.3)	165 (54.1)	14 (4.6)	61 (20.0)	219 (71.8)	25 (8.2)	158 (51.8)	45 (14.8)	102 (33.4)
		χ ² = 7.611; p-value = 0.022			χ ² = 2.689; p-value = 0.261			χ ² = 18.105; p-value = <0.001		

Table 6. Association of practicing pharm D students' profile with their perception and practice of ChatGPT use in clinical training.

of AI course units in their curriculum²⁷. The knowledge gap exist in our study related source of ChatGPT has implications to understand and critically evaluate the generated information for its validity and reliability. Also, these findings strongly recommend to incorporate basics and fundamentals about AI tool use in healthcare academic curriculum for the healthcare students to improve the quality of care and optimizing the patients outcomes²⁸. This recommendation is consistent with UNESCO IITE to implement AI within the curriculums for better perception and wide practice²⁹. Additionally, providing educational interventions regarding AI tools to the practicing HCPs to promote responsible use and integration of these tools in healthcare practice²⁸. A study conducted in India among B. Pharm students revealed that 42.3% of the students had moderate familiarity with AI tool in their curriculum, and 73.1% opined to incorporate AI tool in their academic curriculum³⁰. PCI draft curriculum 2025. Although, many institutions are embraced AI as a core component in the pharmacy education in India, it is worth note to address major concerns that were raised while using AI tools such as data security, accuracy of the information and ethical implications^{26,31}. Additionally, faculty training and encouragement is a pivotal criterion for adoption and integration of the AI technology in pharmacy education³⁰.

Perceived benefits and concerns regarding ChatGPT in clinical training

In our study, though the majority of students agreed ChatGPT can benefit in clinical training, half (50.6%) of the students were neutral, and a 45.1% were have high level perception towards benefits offered by the

ChatGPT. Contrast to our findings, studies conducted among practicing pharmacists in Saudi Arabia (64.1%), Egypt (73.6%), and Jordan (77.2%) revealed high perception towards benefits of ChatGPT in pharmacy practice profession^{18,23,32}. The mixed perception levels observed in our research students directs the presence of uncertainties in responses generated by the ChatGPT in the clinical training. The clinical use of ChatGPT among pharmacy students was low in a study conducted in USA²¹. In general, hesitancy among students to use ChatGPT is likely to be higher than among practicing pharmacists, due to their incomplete training and limited ability to validate the responses generated by ChatGPT. In our study, most students perceived that ChatGPT offers substantial benefits in clinical training, particularly by promoting the rational use of medicines, identifying drug-related problems (DRPs), screening for drug–drug interactions (DDIs), detecting adverse drug reactions (ADRs), calculating doses and dosage regimens, supporting patient counselling, developing pharmaceutical care plans, and documenting clinical pharmacist interventions. These findings are consistent with a study conducted among Pharm D students in Jordan, which also reported that students perceived ChatGPT as competent in assisting with the identification of ADRs, DRPs, and DDIs²². The positive attitude observed in our study is align with the global trends where HCPs express to adopt AI tools in the healthcare practice^{19,33}.

In our study, about 63.3% of the respondents were concerned about ethical issues, such as privacy and confidentiality of patient information, regarding ChatGPT use. Mirror findings (64.0%, 61.2%) were reported in two studies conducted among pharmacy students in Saudi Arabia^{23,26}. Evidence shows that the major limitation of the ChatGPT use is its vulnerability towards cyberattacks and wide spread of the information^{34,35}. While using AI tools in the clinical practice, it is vital to embed the tool to guarantee the privacy, confidentiality, data security, and prevent the misuse of the patient information³¹.

In our study, about two-third of the respondents are concerned about ChatGPT may generate inaccurate or misleading response due to lack of real time verification. Similar findings were also reported in studies conducted among medical and pharmacy students in Jordan, Pharm D students in Jordan, and pharmacy students in Japan^{22,24,25}. A study conducted in India among B. Pharmacy students revealed that 73.8% did not cross-check the information generated by the AI – powered tools with standard resources like textbooks, PubMed search engine, and journals³⁰. A study conducted among healthcare professionals in Saudi Arabia revealed that about two-thirds of them believed that ChatGPT positively contributes to healthcare provision, but they expressed concerns regarding its accuracy and legal issues, suggesting that it requires human supervision³⁶. The concerns raised about accuracy of the information can be nullified by practice of cross-checking and validating the information generated from AI tools with the standard resources. The potential reasons for concerns about data accuracy from ChatGPT may linked with lack of regular update, it works based on previously trained data set, and it follows instructions rather investigation^{22,35}. These inherited gaps present in the learning capabilities of ChatGPT or any other AI tools can result in factual inaccuracies, but it can be posed in the form of scientifically plausible, which is termed as hallucination³⁵. The successful translation of AI models in to real world applications depends not only on the accuracy of the information, but also its ability to generate reliable, safe, and individualized health recommendations³⁷.

More than half of the respondent expressed their concern about the potential risk of bias information by the ChatGPT. Similar findings were replicated in a study conducted by Alghitran et al., among Pharm D students²³. The concern was due to AI trained on non-diverse datasets, and it can affect the correctness of the clinical recommendations. To deal with these issues, it is recommended to train the AI models on diverse and representative data sets, with regular audits to detect and address the inherited biases²³.

Although many students (74.4%) were concerned regarding the use of ChatGPT can reduce the interaction with trainer or mentor or preceptor, only few students (38.2%) were worry about its use can affect the critical thinking and problem-solving skills. Evidence shows that many educators consider it as a threat to the training in acquiring knowledge and problem solving-skills which are essential in healthcare practice³⁸. These findings demonstrate clearly that students want to adopt and integrate the AI tools in their clinical training irrespective of the enhancement of skills or knowledge. In this regard, it is strongly recommended to inculcate the responsible usage AI tools among faculty and students by providing training regarding use of these tools, extent of use, validation of the information retrieved, and challenges faced to use AI tools in clinical training specifically connected with clinical pharmacy activities.

An interesting finding is that more than half (64.1%) of the students were concerned that using ChatGPT could increase the similarity index and AI detection score in their reports. In contrast, only 40.5% expressed concern about the underutilization of standard information resources in the clinical pharmacy field. About 51.6% of the Pharm D students agreed ChatGPT helps equally as standard resources (UpToDate, Medscape, Lexicomp) in the Jordan study²². Any response to a clinical query posed to ChatGPT requires validation against information from standard resources or expert opinion to ensure its correctness and applicability. Therefore, ChatGPT or any other AI tool should not be considered a substitute for standard resources or human interpretation of these resources for personalized application in healthcare. These findings and interpretations recommend strengthening educational regulations related to similarity indices and AI detection scores to enhance the appropriate use of standard resources under supervisor guidance. This approach can help students develop the skills and competencies needed to handle real-world clinical practice and limit academic misconduct. Meanwhile, it is worth noting that incorporating AI tools and their applications into the curriculum and training staff in these areas, can promote the responsible use of AI tools in clinical practice.

Practice of ChatGPT in clinical training

In our study, more than half of the students showed high practice and few had moderate practice of ChatGPT in their clinical training with diverse extent across the different clinical pharmacy domains. Majority (83.5) of the students are using ChatGPT in getting counselling tips and providing effective patient counselling and communication to the patients. Similar findings were mirrored among students (81.9%) of Jordan indicating

that ChatGPT is a beneficial learning tool for counselling and mental health education³⁹. Contrast to our findings, only 27% of the pharmacy students are using ChatGPT in generating counselling tips for the patient education in the United States of America²¹. An evidence supports that AI-based simulations can increase the interactions of medical students with the patients⁴⁰. A mixed-methods study involving 27 medical students from three UK universities demonstrated a significant improvement in self-reported confidence across all communication domains following the ChatGPT simulation by using advanced voice mode feature⁴¹. These finding strongly suggest to incorporate ChatGPT and other LLM in training of the pharmacy students to acquire communication skills and deliver the organized evidence based educational content to the patients. An exploratory case study conducted on ChatGPT in delivering instructions to the patients about medications revealed that the instructions are potential to cause harm and non-adherence⁴². Additionally, the evaluators reported ChatGPT provides correct responses to drugs related information but it is not complete⁴². Contrast to the exploratory study findings, a study conducted by Huang et al., reported that ChatGPT excelled in drug counselling, achieving nearly equivalent score to the clinical pharmacist¹⁴. These evidences clearly demonstrates that ChatGPT guides the trainers or trainee to enhance their communication skills for implementing effective counselling sessions to the patients, but not as a reliable standalone source of patient drug information.

In our study, about 70.6% of the respondents were utilizing ChatGPT for screening and identification of the drug-drug interactions. These findings resembles with the Jordan study conducted among Pharm D students²². Here, it is critical to address the standards of ChatGPT in providing drug related responses. A study demonstrated and revealed that ChatGPT generated either wrong or partly wrong answers for most 50-drug related questions⁴³. Building on this, a study on drug-herbal interaction evaluation for traditional Chinese medicine using ChatGPT revealed inappropriate and lack of information to provide patient recommendations⁴⁴. A study demonstrated that ChatGPT show lower sensitivity in to predicting DDIs in real world scenarios compared with pharmacist evaluation⁴⁵. Across different AI Chatbots, Bing AI showed highest accuracy and specificity over the Google's Bard, ChatGPT-3.5, and ChatGPT-4 in predicting DDIs¹⁹. Also, the same study findings revealed ChatGPT 3.5, and 4 shown high variability in accuracy while examining two different drug classes¹⁹. Based on the reported evidence regarding predicting DDIs, it was strongly recommended to the PharmD students or clinical pharmacy practitioners to validate the responses generated by AI tools with standard references like Micromedex, Drugs.com, and UpToDate.

Findings of our study revealed that about 73.4% of the PharmD students were using ChatGPT to respond to drug or poison information queries from the healthcare staff or students in their clinical training. These findings clearly demonstrate to us the high dependence of the students on AI models to respond to clinical queries in their clinical training. However, a comparative study conducted on ChatGPT-3.5 and ChatGPT-4 against Lexicomp yielded limited accuracy and reproducibility in their responses⁴⁶. The authors strongly recommended that healthcare professionals be cautious while using ChatGPT for the purpose of retrieval of drug information⁴⁶. Despite, ChatGPT operation being user-friendly and accessing quick responses by posing word or voice text, it doesn't replace the evidence-based databases such as Lexicomp[®], Micromedex, or UpToDate. Educators or trainers must appraise the information and verify the source to prevent false information; this can promote steps towards safe application of the AI models in clinical decision-making.

More than half (58.5%) of the students were utilizing ChatGPT to calculate medication doses and dosing regimens for the patients during their clinical training. Evidence indicates that ChatGPT responses were correct only for 35% of drug-dosing-related questions⁴⁷. Calculating dosage is a complex task that necessitates precise and context-specific information⁴⁷. The complexities involved in dosing calculations based on patient-specific characteristics also contribute to ChatGPT's underperformance with dosing-related queries⁴⁷. Although ChatGPT usage is low in dose calculation compared with other clinical pharmacy activities, still a considerable proportion of the PharmD students are dependent on this AI tool irrespective of the validity of dosing information. It was strongly recommended to develop more sophisticated programs that are integrated with real-world patient care data, which can improve the model's performance in dose calculation⁴⁷. Additionally, PharmD students must perform a validity check with standard resources or expert opinion before applying the ChatGPT-generated information in patient care to avoid drug-related consequences.

Our study found that 70.4% of the respondents were using ChatGPT to identify and manage the ADRs in their clinical training. In contrast with our findings, a study conducted among Egyptian pharmacists shows that only 35.5% were utilizing ChatGPT in the identification and management of the ADRs³². The variations in knowledge, skills, and expertise observed among training and practicing pharmacists justified the disparities between the studies. Huang et al., reported that ChatGPT was less accurate and less complete compared with clinical pharmacists in identifying ADRs and performing causality assessment for the reported ADRs¹⁴. Identification of ADRs requires integration of structured (patient demographics, lab reports) and unstructured (clinical notes) data in real-world clinical scenarios by using advanced deep learning and NLP methods⁴⁸. These methods help in predicting factors that influence the likelihood of ADRs, which offers insights for the targeted monitoring and mitigation strategies⁴⁸. These findings demonstrate that ChatGPT or any other LLM is for the identification of the ADRs, not a substitute for the human pharmacists' role in the pharmacovigilance sector.

Nearly two-thirds of the respondents in our study reported using ChatGPT to select a suitable drug for disease management. In contrast, a study conducted in the USA found that only 20% used it to select drug treatments and 27% used it to choose non-drug treatments for patient management²¹. A study among Egyptian pharmacists showed that 30.6% were using ChatGPT as a drug or disease information resource³². The high use of ChatGPT for retrieving information related to the selection of drugs in patient care and also for providing different clinical pharmacy services suggests that academic institutions and universities should provide guidelines or regulations for the responsible use of AI tools in academic and clinical practice in India by considering evidence from the high-income countries. A report published by Academic Medicine provides some guidance regarding competencies required for the healthcare professionals, including pharmacists, who are using AI tools in the

healthcare system⁴⁹. This group is a multidisciplinary team that formulated six clinical competency statements that guide academic programs to help their stakeholders with the rational use of AI tools in healthcare. These six competencies include (1) Basic knowledge about AI and its application in the healthcare sector; (2) Considering the influence of social, economic, and political systems on AI-based tools and their impact on justice, equity, and ethics; (3) Carrying out AI-enhanced clinical encounters that integrate diverse sources of information in creating patient-centred care plans; (4) Evidence-based evaluation of the AI tools for their quality, accuracy, safety, and contextual appropriateness. The fourth competency also addresses biases resulting from AI-based tools and their underlying patient or population data sets; (5) Workflow analysis (change in team, roles, and responsibilities) for the implementation of AI-based tools in healthcare practice; (6) Participating in continuous professional development and practice-based improvement activities related to AI tools in healthcare.

Variables such as previous experience using AI tools and prior training on AI tool use were significantly associated with greater practice of ChatGPT. Conversely, the perceived benefits of ChatGPT were significantly correlated with students who had completed internships and received prior training on AI tools. In our study significant association conveys that students are confident to practice with ChatGPT based on their previous experience or training on AI tool use. Generally, receiving training or gaining experience with AI tools can enhance the ability to evaluate retrieved information in light of standard resources. In contrast with our findings, a study conducted in Egypt revealed that prior understanding of AI technology doesn't correlate with perception, concerns, and practice of ChatGPT among practicing pharmacists³².

The respondents perceived concerns about ChatGPT in clinical training significantly positively weakly correlated with their knowledge score. There was a moderate positive correlation between the perceived benefits score and the practices of ChatGPT use in clinical training, while the perceived concerns score showed a weakly negative correlation. In contrast with our findings, there were no correlations between perceptions, concerns, and practices of ChatGPT use among the Egyptian pharmacists³². In contrast, a study conducted in Jordan among PharmD students demonstrated a connection between perception and concern levels²². It can influence the ChatGPT use or its recommendation among student community. Practicing pharmacists may have high clinical reasoning and judgment and rely on the information from validated resources⁵⁰, which can lead to less variability in their perception or practices regarding ChatGPT use. Additionally, our study findings offer insights for AI training programs aimed at promoting a rational approach to the benefits and challenges of using AI tools, based on available literature evidence regarding ChatGPT's performance in clinical pharmacy. Understanding benefits and concerns about ChatGPT use in clinical pharmacy activities in light of available evidence can enhance the Pharm D students or practicing pharmacists' responsible practice of ChatGPT.

Strengths and limitations

This is the first study conducted in India to evaluate the knowledge, perception, and practice of ChatGPT in clinical training among PharmD students. Researchers/academicians/practitioners need to consider the following study limitations while interpreting our study findings towards making policies, developing clinical practice guidelines, and integrating AI tools into healthcare practice. Firstly, we used a non-probable, snowball sampling technique to capture the required samples that can introduce the selection bias, and respondents may not accurately represent the entire population. Secondly, the online-based survey limited us to calculating the response rate of the participants. Thirdly, a self-administered mode of data collection can also introduce bias, as students may have provided information that is not entirely accurate or reflective of their true beliefs and intentions regarding ChatGPT in their clinical training. Fourthly, we didn't address a few aspects, like why students are not using ChatGPT for their clinical training, how often they use ChatGPT, what their satisfaction level is upon use of ChatGPT, and whether they validated the accuracy of the response given by ChatGPT with their knowledge or standard resources. Fifthly, we didn't evaluate the impact of ChatGPT on the clinical decision or patient outcomes. Finally, despite the large sample size ($n = 407$) drawn from 49 institutions across 12 states, the predominance of respondents from Andhra Pradesh (72.7%) may limit the generalizability of our study's findings. The limitations observed in our study promptly recommend the need for a mixed-method evaluation of the opinions of the healthcare practitioners regarding the AI tool's performance in healthcare practice. Furthermore, we only focused on ChatGPT, which is one of the many LLMs. Future research is expected to do a comparative analysis of the various LLMs in terms of accuracy, completeness, reproducibility, correctness, and reliability of providing information related to performing diverse clinical pharmacy activities. However, our research provides standalone evidence for educational institutions, clinical pharmacy practitioners, healthcare professionals, academic regulators, and researchers to understand students' perceptions and practices regarding the application of ChatGPT in Indian clinical training settings. Although the present study was conducted prior to the release of **ChatGPT-5**, the findings remain relevant in understanding how students adapt to rapidly advancing AI technologies in clinical training.

Future studies can be planned to address the limitations observed in our research and strengthen the quality and reliability of the study tool. These include the inclusion of the psychometric expert in the evaluation of the content, cognitive interview of the students before deployment of the research tool, and performing exploratory or confirmatory factor analysis to improve the internal coherence of the domains. Our study also recommends utilizing Likert scale responses to minimize the overestimation of practices.

Conclusion

Our study concludes that the majority of the respondents have good knowledge regarding ChatGPT. About half of the respondents had a neutral or high perception of the benefits offered by ChatGPT, with varying opinions at the individual activity level. The majority of the respondents were concern about ChatGPT use can have impact on the interaction with the preceptor. More than half of the respondents were practicing using ChatGPT in the clinical pharmacy training, with variations at the individual activity level. In practice, it was observed that

the majority of the PharmD students are using ChatGPT for general tasks where individualization or tailoring of information is not necessary, and the response is not influenced by the patient characteristics. However, low levels of usage were observed for tasks such as dose calculation and selecting the medicine based on the patient's condition. The variations observed in the practices of PharmD students are due to underlying concerns about ChatGPT responses, such as privacy and confidentiality of patient information, accuracy, and misleading information; biased responses; limited scope and applicability; and reduced interaction with mentors. In line with other literature findings, pharmacy students from around the world indicate that they have used ChatGPT and they are eager to learn more about AI and its integration into their professional practices. These findings strongly recommend education regulators incorporate AI tools and their applications into the curriculum and train staff in these areas, which can promote the responsible use of AI tools in clinical practice. Educators or trainers must appraise the information and verify the source to prevent false information; this can promote steps towards safe application of the AI models in clinical decision-making.

Data availability

The data that support the findings of this study are available from the corresponding author, Narayana Goruntla, upon reasonable request.

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Declarations

Competing interests

The authors declare no competing interests.

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